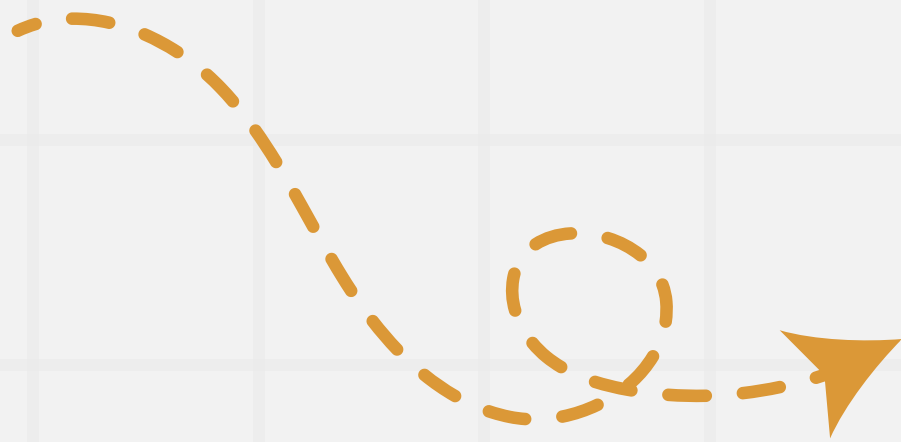




THE COMPLETE REACTJS FRONTEND INTERVIEW GUIDE



decodeWithSaurav 🔍



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01

ReactJs Questions

- 1 What's new in React 19, and how does it enhance concurrent rendering?
- 2 Explain how the React Compiler automatically optimizes hooks and re-renders.
- 3 How do Server Actions differ from traditional REST or GraphQL data fetching?
- 4 Explain Transition APIs and how they improve UI responsiveness.
- 5 How does React Suspense simplify async rendering and data loading?
- 6 What is the use() hook, and how does it integrate async behavior in components?
- 7 Describe the difference between Server Components and Client Components.
- 8 How do error boundaries and React.lazy work together in React 19?
- 9 How does React handle automatic batching and why is it critical for scalability?
- 10 What's the future of React Server Components in large-scale web apps?

02

Performance & Optimization Questions

- 1 What are Web Workers, and how do they improve app responsiveness?
- 2 How do you offload CPU-heavy computations in React using Workers?
- 3 What's the difference between Dedicated, Shared, and Service Workers?
- 4 How does OffscreenCanvas improve visual rendering performance?
- 5 When would you use WebAssembly (WASM) in frontend optimization?
- 6 How do you profile React performance using React DevTools Profiler?
- 7 How do you reduce bundle size using code-splitting and tree shaking?
- 8 What's the purpose of React.memo and useMemo for performance tuning?
- 9 How do you prevent unnecessary re-renders in React components?
- 10 How do you implement lazy hydration for faster Time to Interactive (TTI)?

03

Frontend System Design Questions

- 1 How do you design a scalable React microfrontend architecture?
- 2 How do you build and manage a shared design system across multiple apps?
- 3 What's your strategy for global state management (Redux vs Zustand vs Context)?
- 4 How would you structure a monorepo for multiple React applications?
- 5 How do you implement CI/CD pipelines for frontend deployments?
- 6 How do you ensure accessibility (a11y) compliance in large React apps?
- 7 What is your approach to feature flags and A/B testing in React?
- 8 How would you design a progressive web app (PWA) using React?
- 9 What's your approach for optimistic UI updates in React Query or SWR?
- 10 How do you ensure observability and error tracking in production React apps?

04

ReactJS Frontend Coding Challenges

- 1 Build a debounce + throttle utility and integrate it with a search bar.
- 2 Create a custom useFetch hook with retry, cancel, and loading states.
- 3 Build a drag-and-drop Kanban board using React DnD.
- 4 Implement infinite scrolling using the Intersection Observer API.
- 5 Design a responsive Masonry layout with pure CSS Grid and React.
- 6 Create skeleton screens and lazy loading for image-heavy components.
- 7 Build a dark/light theme switcher using Context API.
- 8 Implement a modal manager system using React Portals.
- 9 Develop a real-time chat interface using WebSockets or Firebase.
- 10 Implement a multi-file uploader with drag, preview, and progress tracking.

05

JavaScript / Browser Internals Questions

- 1 Explain structured cloning and its role in data transfer for Web Workers.
- 2 What's the difference between microtasks and macrotasks in JS event loop?
- 3 How does the call stack, heap, and queue work in modern browsers?
- 4 Explain how memory leaks occur in React apps and how to fix them.
- 5 How does import maps change module resolution in browsers?
- 6 Explain shadow DOM and its relevance in modern component architectures.
- 7 How do service workers handle caching and offline states?
- 8 What are performance APIs (Navigation, Paint, Long Tasks) used for?
- 9 How do event delegation and propagation affect React's synthetic events?
- 10 How does concurrent rendering interact with the browser event loop?

06

React Architecture Questions

- 1 How do you architect a modular React monolith for large-scale apps?
- 2 Explain feature-based folder structures for maintainability.
- 3 How do you ensure isolation in microfrontends with shared dependencies?
- 4 What are component composition patterns, and how do they differ from HOCs?
- 5 Explain Container-Presenter pattern — when is it useful?
- 6 How would you migrate a legacy React app to TypeScript safely?
- 7 How do you structure design systems (Figma → React) for scalability?
- 8 How do you handle API versioning in frontend codebases?
- 9 How do you design for offline-first experiences in React?
- 10 When should you use context modules vs global state libraries (Zustand/Recoil)?

07

Performance & Debugging Deep Dives

- 1 How do you diagnose memory leaks in a React SPA?
- 2 How do you use React Profiler and Chrome Performance tools together?
- 3 How do you debug re-render chains caused by prop changes?
- 4 How do you optimize React's reconciliation for large lists?
- 5 How do you fix slow hydration in SSR apps?
- 6 How do you optimize Lighthouse metrics (TTFB, FCP, TTI)?
- 7 How do you manage code-splitting at route and component levels?
- 8 How do you detect network waterfall issues in React?
- 9 What's your method for stale closures in async functions?
- 10 How do you prevent unnecessary state updates with custom hooks?

08

Fullstack & Integration Awareness

- 1 How do you design React apps that consume GraphQL / REST / gRPC efficiently?
- 2 How do you architect API caching layers using service workers or IndexedDB?
- 3 How do you handle JWT refresh flow securely in SPAs?
- 4 Explain API error handling frameworks you've designed.
- 5 How do you structure a Backend-for-Frontend (BFF) for React apps?
- 6 How do you implement CI/CD pipelines for production-grade React apps?
- 7 How do you handle feature rollout with canary deployments?
- 8 How do you monitor frontend errors (Sentry, Datadog)?
- 9 How do you ensure data consistency across distributed frontend services?
- 10 What's your view on server-driven UI (SDUI) in React ecosystems?

- 1 How do you prevent XSS, CSRF, and clickjacking in React apps?
- 2 How do you implement Content Security Policy (CSP)?
- 3 How do you sanitize user input in React forms?
- 4 What tools do you use for accessibility testing (axe, Lighthouse)?
- 5 How do you ensure keyboard and screen-reader compatibility?
- 6 How do you handle sensitive data in frontend securely?
- 7 How do you implement error masking in production?
- 8 How do you manage session timeouts gracefully?
- 9 How do you secure cookies and auth tokens?
- 10 How do you apply OWASP Top 10 principles in frontend apps?

10

Behavioral & Leadership Questions

- 1 How do you mentor junior developers on frontend best practices?
- 2 How do you lead architecture reviews effectively?
- 3 How do you balance tech debt vs delivery in sprints?
- 4 How do you define and enforce coding standards across teams?
- 5 How do you collaborate with design and product teams?
- 6 How do you communicate frontend performance metrics to non-engineers?
- 7 How do you introduce new libraries or frameworks safely?
- 8 How do you prioritize refactors in large systems?
- 9 What's your strategy for feature flag rollbacks?
- 10 How do you balance innovation vs consistency in frontend systems?